BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE423: Computer Graphics

Assignment 01 | Total Marks: 20 | Summer 2025

Submission: July 09, 2025

Q1. A video runs at 3840×2160 resolution, 60 FPS, with 24-bit color depth. The GPU processes 95,000 pixels/ms. Each frame uses 24,883,200 bytes in memory.

a. Find the number of pixels per second and total bits per second processed. [2]

b. Can the GPU render one full frame (3840×2160) within 16.67 ms? Show your calculation. [4]

c. Using the 24-bit depth and 24,883,200-byte frame size, calculate how many pixels per frame are stored. Compare it with 3840×2160. [4]

Q2. **a.** Using the **DDA** (Digital Differential Analyzer) line drawing algorithm, compute all the pixels for a line segment from (5, -12) to (0, -5). Show all the steps at each stage. [4]

b. An autonomous delivery robot is stationed at a point where the line $7x - 4y + 56 = 0$ intersects the y-axis on a city grid. The robot must travel in a straight line to deliver a package to the coordinates (10, 5).

Using the **Midpoint** line drawing algorithm, compute the first 7 pixels of the original zone. Show the current value of d at each step as it gets updated. **Apply 8 way symmetry** (Zone conversion) if required. [6]

Submission Form: https://forms.gle/bFjgThrmfud5XxoV6